United States Patent	12403266
Kind Code	B1
Date of Patent	September 02, 2025
Inventor(s)	Denyer; Timothy et al.

Medicament delivery device

Abstract

A medicament delivery device has a needle, a body and a needle cover which is axially movable from an initial position to a holding position to a locked position in which the needle cover covers the distal end of the needle. The device has a biasing member configured to bias the needle cover distally, and a guide member configured to rotate relative to the needle cover. One of the guide member and the needle cover has a pin and the other of the guide member and the needle cover has a guide path for guiding the pin. The guide path has a first ramp, a second ramp and a locking abutment surface. When the needle cover is in the locked position then the locking abutment surface is configured to engage the pin to prevent the needle cover from moving proximally away from the locked position.

Inventors: **Denyer; Timothy (Melbourn, GB), Bradford; James (Melbourn, GB), Hee-Hanson; Alexander (Melbourn, GB), Wilson; Robert (Melbourn, GB), Twite; Dean (Melbourn, GB), Lever; Thomas (Melbourn, GB)**

Applicant: **Genzyme Corporation** (Cambridge, MA)

Family ID: **96882049**

Assignee: **Genzyme Corporation (Cambridge, MA)**

Appl. No.: **18/819371**

Filed: **August 29, 2024**

Publication Classification

Int. Cl.: **A61M5/32** (20060101); **A61M5/20** (20060101)

U.S. Cl.:

CPC **A61M5/322** (20130101); **A61M5/20** (20130101); **A61M5/3272** (20130101); A61M2005/3267 (20130101); A61M5/3271 (20130101)

Field of Classification Search

CPC: A61M (5/326); A61M (5/3271); A61M (2005/3267); A61M (5/6272)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4118611	12/1977	Harris	N/A	N/A
4994045	12/1990	Ranford	604/263	A61M 5/3271
6547764	12/2002	Larsen et al.	N/A	N/A
8858510	12/2013	Karlsson	604/198	A61M 5/3272
9919107	12/2017	Imai	N/A	A61M 5/326
12329952	12/2024	Denyer et al.	N/A	N/A
2002/0133122	12/2001	Giambattista et al.	N/A	N/A
2007/0078408	12/2006	Wang	604/110	A61M 5/24
2012/0041368	12/2011	Karlsson	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 2022/117682	12/2021	WO	N/A

OTHER PUBLICATIONS

Needle-based injection systems for medical use requirements and test methods, Part 1: Needle injection systems, ISO 11608 1:2014(E), Third Edition, Switzerland, ISO, Dec. 15, 2014, pp. 1-13. cited by applicant

Speciale et al., “Snap-Through Buckling Mechanism for Frequency-up Conversion in Piezoelectric Energy Harvesting,” Applied Sciences, May 23, 2020, 10(10):3614, 18 pages. cited by applicant

U.S. Appl. No. 18/819,222, filed Aug. 29, 2024, Timothy Denyer. cited by applicant

U.S. Appl. No. 18/819,040, filed Aug. 29, 2024, Timothy Denyer. cited by applicant

Primary Examiner: Bouchelle; Laura A

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

TECHNICAL FIELD

(1) This application relates to a medicament delivery device and a method of using a medicament delivery device.

BACKGROUND

(2) Medicament delivery devices are used to deliver a range of medicaments.

(3) In some devices, the device must be held in a holding position at an injection site to ensure that the correct dose of medicament is dispensed from the device, before removing the device from the

injection site.
(4) It may be difficult to hold the device in the holding position whilst the medicament is dispensed. This may result in pain, discomfort, a wet injection site, early device removal and/or partial delivery of the medicament.
(5) It is an aspect of the present disclosure to provide an improved medicament delivery device.
SUMMARY
(6) According to a first aspect of the present disclosure, there is provided a medicament delivery device comprising a needle for injecting medicament; a body having a proximal end and a distal end; a needle cover, wherein the needle cover is proximally movable relative to the body from an initial position to a holding position in which the needle protrudes from the distal end of the needle cover for injecting medicament, and wherein the needle cover is distally movable relative to the body from the holding position to a locked position in which the needle cover covers the distal end of the needle; a biasing member configured to bias the needle cover distally; and a guide member configured to rotate relative to the needle cover, wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path for guiding the pin, wherein the guide path comprises a first ramp, a second ramp and a locking abutment surface, wherein the pin is positioned at a first position when the needle cover is in the initial position, and wherein the first ramp is configured to engage the pin when the needle cover moves axially from the initial position towards the holding position for rotating the guide member relative to the needle cover, wherein the pin is positioned at a second position when the needle cover is in the holding position, wherein the second ramp is configured to engage the pin when the needle cover moves axially from the holding position towards the locked position for rotating the guide member relative to the needle cover, wherein the pin is positioned at a third position when the needle cover is in the locked position, wherein the locking abutment surface is configured to engage the pin to prevent the needle cover from moving proximally away from the locked position.
(7) The locking abutment surface may comprise a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device.
(8) When the pin is in the second position it may engage the first ramp.
(9) The first position may be in the guide path. The guide path may comprise an axially extending portion, wherein the first position is in the axially extending portion. The axially extending portion may be parallel with the longitudinal axis of the medicament delivery device.
(10) The needle cover may be rotationally fixed relative to the body. The guide member may be axially fixed relative to the body.
(11) The first ramp may comprise a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device. The second ramp may comprise a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device.
(12) The body may be configured to be gripped by a user.
(13) The needle cover may comprise the pin and the guide member may comprise the guide path. In the initial position the needle cover may cover the distal end of the needle.
(14) The guide member may comprise an outer surface, and wherein the guide path is recessed within the outer surface.
(15) The needle cover may comprise an arm, wherein the pin is provided on the arm. The pin may be provided on the free end of the arm.
(16) The first ramp may be configured to rotate the guide member in a first direction when the needle cover moves axially away from the initial position towards the holding position. The second ramp may be configured to rotate the guide member in the first direction when the needle cover moves axially from the holding position towards the locked position. The locking abutment surface may be configured to prevent rotation of the guide member in a second direction which is counter to the first direction when the pin is in the third position.

(17) The medicament delivery device may be configured to inject greater than 2 ml of medicament. The medicament delivery device may be configured to inject medicament having a viscosity of greater than 25 cP.
(18) The device may comprise a mechanism which automatically causes medicament to be dispensed from the device, wherein when the needle cover moves from the initial position towards the holding position it releases the mechanism and medicament is automatically dispensed from the device.
(19) The device may comprise the medicament.
(20) According to another aspect of the present disclosure, there is provided a method of locking a medicament delivery device after medicament has been dispensed from the medicament delivery device, the method comprising moving a needle cover of the device to a locked position in which the needle cover covers the distal end of a needle of the device, wherein the device comprises a guide member configured to rotate relative to the needle cover, and wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path for guiding the pin, wherein the guide path comprises a locking abutment surface, and wherein when the needle cover is in the locked position then proximal movement of the needle cover away from the locked position is blocked by the pin engaging the locking abutment surface.
(21) The medicament delivery device may have any of the features as described and/or contemplated herein.
(22) According to another aspect of the present disclosure, there is provided a method of using a medicament delivery device, the method comprising removing the medicament delivery device from an injection site, wherein removing the medicament delivery device from the injection site causes a needle cover of the device to move to a locked position in which the needle cover covers the distal end of a needle of the device, wherein the device comprises a guide member configured to rotate relative to the needle cover, and wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path comprising a locking abutment surface, and wherein when the needle cover is in the locked position then proximal movement of the needle cover away from the locked position is blocked by the pin engaging the locking abutment surface.
(23) The method may further comprise the preceding step of pressing the medicament delivery device against the injection site to move the needle cover from an initial position to a holding position for dispensing medicament from the device.
(24) The method may further comprise holding the medicament delivery device in the holding position whilst medicament is dispensed from the device.
(25) The guide path may further comprise a first ramp which engages the pin when the needle cover moves axially from the initial position towards the holding position for rotating the guide member relative to the needle cover.
(26) The guide path may further comprise a second ramp which engages the pin when the needle cover moves from the holding position towards the locked position for rotating the guide member relative to the needle cover.
(27) The movement of the needle cover from the initial position to the holding position may cause medicament to be dispensed from the device via the needle.
(28) The device may comprise a mechanism configured to dispense medicament from the device via the needle when the needle cover reaches a predetermined axial position. Movement of the needle cover from the initial position to the holding position may trigger the mechanism to dispense medicament from the device via the needle.
(29) The medicament delivery device may comprise a container for containing the medicament. The medicament may be located in the container. The container may be a syringe. The syringe may

comprise the needle. The container may be a cartridge which is initially separated from the needle when the needle cover is in the initial position.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Embodiments will now be described, by way of example only, with reference to the accompanying drawings, in which:

(2) FIG. **1**A shows an injector device with a cap attached;

(3) FIG. **1**B shows the injector device of FIG. **1**A with the cap removed;

(4) FIG. **2** shows a medicament delivery device;

(5) FIG. **3**A shows a simplified view of an example medicament delivery device when the needle cover is in the initial position;

(6) FIG. **3**B shows a view of the guide path of the medicament delivery device of FIG. **3**A, indicating the position of the pin when the needle cover is in the initial position;

(7) FIG. **4**A shows a simplified view of the medicament delivery device of FIG. **3**A when the needle cover is in the holding position;

(8) FIG. **4**B shows a view of the guide path of the medicament delivery device of FIG. **3**A, indicating the position of the pin when the needle cover is in the holding position;

(9) FIG. **5**A shows a simplified view of the medicament delivery device of FIG. **3**A when the needle cover is in the locked position;

(10) FIG. **5**B shows a view of the guide path of the medicament delivery device of FIG. **3**A, indicating the position of the pin when the needle cover is in the locked position; and

(11) FIG. **6** shows a perspective view of a guide path for a medicament delivery device.

DETAILED DESCRIPTION

(12) A drug delivery device, as described herein, may be configured to inject a medicament into a patient. For example, delivery could be sub-cutaneous, intra-muscular, or intravenous. Such a device could be operated by a patient or care-giver, such as a nurse or physician, and can include various types of safety syringe, pen-injector, or auto-injector. The device can include a cartridge-based system that requires piercing a sealed ampule before use. Volumes of medicament delivered with these various devices can range from about 0.5 ml to about 2 ml. Yet another device can include a large volume device (“LVD”) or patch pump, configured to adhere to a patient's skin for a period of time (e.g., about 5, 15, 30, 60, or 120 minutes) to deliver a “large” volume of medicament (typically about 2 ml to about 10 ml).

(13) In combination with a specific medicament, the presently described devices may also be customized in order to operate within required specifications. For example, the device may be customized to inject a medicament within a certain time period (e.g., about 3 to about 20 seconds for auto-injectors, and about 10 minutes to about 60 minutes for an LVD). Other specifications can include a low or minimal level of discomfort, or to certain conditions related to human factors, shelf-life, expiry, biocompatibility, environmental considerations, etc. Such variations can arise due to various factors, such as, for example, a drug ranging in viscosity from about 3 cP to about 50 cP. Consequently, a drug delivery device will often include a hollow needle ranging from about 25 to about 31 Gauge in size. Common sizes are 27 and 29 Gauge.

(14) The delivery devices described herein can also include one or more automated functions. For example, one or more of needle insertion, medicament injection, and needle retraction can be automated. Energy for one or more automation steps can be provided by one or more energy sources. Energy sources can include, for example, mechanical, pneumatic, chemical, or electrical energy. For example, mechanical energy sources can include springs, levers, elastomers, or other mechanical mechanisms to store or release energy. One or more energy sources can be combined

into a single device. Devices can further include gears, valves, or other mechanisms to convert energy into movement of one or more components of a device.

(15) The one or more automated functions of an auto-injector may each be activated via an activation mechanism. Such an activation mechanism can include one or more of a button, a lever, a needle sleeve, or other activation component. Activation of an automated function may be a one-step or multi-step process. That is, a user may need to activate one or more activation components in order to cause the automated function. For example, in a one-step process, a user may depress a needle sleeve against their body in order to cause injection of a medicament. Other devices may require a multi-step activation of an automated function. For example, a user may be required to depress a button and retract a needle shield in order to cause injection.

(16) In addition, activation of one automated function may activate one or more subsequent automated functions, thereby forming an activation sequence. For example, activation of a first automated function may activate at least two of needle insertion, medicament injection, and needle retraction. Some devices may also require a specific sequence of steps to cause the one or more automated functions to occur. Other devices may operate with a sequence of independent steps.

(17) Some delivery devices can include one or more functions of a safety syringe, pen-injector, or auto-injector. For example, a delivery device could include a mechanical energy source configured to automatically inject a medicament (as typically found in an auto-injector) and a dose setting mechanism (as typically found in a pen-injector).

(18) According to some embodiments of the present disclosure, an exemplary drug delivery device **10** is shown in FIGS. **1**A & **1**B. Device **10**, as described above, is configured to inject a medicament into a patient's body. Device **10** includes a housing **11** which typically contains a reservoir containing the medicament to be injected (e.g., a syringe) and the components required to facilitate one or more steps of the delivery process. Device **10** can also include a cap assembly **12** that can be detachably mounted to the housing **11**. Typically a user must remove cap **12** from housing **11** before device **10** can be operated.

(19) As shown, housing **11** is substantially cylindrical and has a substantially constant diameter along the longitudinal axis X. The housing **11** has a distal region **20** and a proximal region **21**. The term “distal” refers to a location that is relatively closer to a site of injection, and the term “proximal” refers to a location that is relatively further away from the injection site.

(20) Device **10** can also include a needle sleeve **13** coupled to housing **11** to permit movement of sleeve **13** relative to housing **11**. For example, sleeve **13** can move in a longitudinal direction parallel to longitudinal axis X. Specifically, movement of sleeve **13** in a proximal direction can permit a needle **17** to extend from distal region **20** of housing **11**.

(21) Insertion of needle **17** can occur via several mechanisms. For example, needle **17** may be fixedly located relative to housing **11** and initially be located within an extended needle sleeve **13**. Proximal movement of sleeve **13** by placing a distal end of sleeve **13** against a patient's body and moving housing **11** in a distal direction will uncover the distal end of needle **17**. Such relative movement allows the distal end of needle **17** to extend into the patient's body. Such insertion is termed “manual” insertion as needle **17** is manually inserted via the patient's manual movement of housing **11** relative to sleeve **13**.

(22) Another form of insertion is “automated,” whereby needle **17** moves relative to housing **11**. Such insertion can be triggered by movement of sleeve **13** or by another form of activation, such as, for example, a button **22**. As shown in FIGS. **1**A & **1**B, button **22** is located at a proximal end of housing **11**. However, in other embodiments, button **22** could be located on a side of housing **11**.

(23) Other manual or automated features can include drug injection or needle retraction, or both. Injection is the process by which a bung or piston **23** is moved from a proximal location within a syringe (not shown) to a more distal location within the syringe in order to force a medicament from the syringe through needle **17**. In some embodiments, a drive spring (not shown) is under compression before device **10** is activated. A proximal end of the drive spring can be fixed within

proximal region **21** of housing **11**, and a distal end of the drive spring can be configured to apply a compressive force to a proximal surface of piston **23**. Following activation, at least part of the energy stored in the drive spring can be applied to the proximal surface of piston **23**. This compressive force can act on piston **23** to move it in a distal direction. Such distal movement acts to compress the liquid medicament within the syringe, forcing it out of needle **17**.

(24) Following injection, needle **17** can be retracted within sleeve **13** or housing **11**. Retraction can occur when sleeve **13** moves distally as a user removes device **10** from a patient's body. This can occur as needle **17** remains fixedly located relative to housing **11**. Once a distal end of sleeve **13** has moved past a distal end of needle **17**, and needle **17** is covered, sleeve **13** can be locked. Such locking can include locking any proximal movement of sleeve **13** relative to housing **11**. Another form of needle retraction can occur if needle **17** is moved relative to housing **11**. Such movement can occur if the syringe within housing **11** is moved in a proximal direction relative to housing **11**. This proximal movement can be achieved by using a retraction spring (not shown), located in distal region **20**. A compressed retraction spring, when activated, can supply sufficient force to the syringe to move it in a proximal direction. Following sufficient retraction, any relative movement between needle **17** and housing **11** can be locked with a locking mechanism. In addition, button **22** or other components of device **10** can be locked as required.

(25) FIG. **2** shows a simplified view of a medicament delivery device **100**. The medicament delivery device **100** has a needle **117** for injecting medicament, a body **111** having a proximal end **130** and a distal end **131**, and a needle cover **113**. The needle **117** has a distal end **140**. The needle cover **113** is proximally movable relative to the body **111** between an initial position, in which the needle cover **113** covers the distal end **140** of the needle, and a holding position for dispensing medicament from the device. The device **100** extends along an axis **144**.

(26) The device **100** is shown in the initial position in FIG. **2**. In the holding position the needle cover **113** is located proximally relative to the initial position. In the holding position the needle **117** protrudes from the distal end **120** of the needle cover **113**.

(27) The medicament delivery device further comprises a biasing member such as a spring **118** configured to bias the needle cover **113** axially in the distal direction. The distal direction is indicated by the direction of the arrow **142** in FIG. **2**.

(28) The medicament delivery device has a plunger **121** which is axially movable within a syringe **123** of the device to dispense medicament from the syringe **123** via the needle **117**.

(29) The medicament delivery device has a collar **119**. The collar **119** is axially fixed relative to the body **111**. The collar **119** interfaces with the plunger **121** via a screw thread **122**. The medicament delivery device **100** has a biasing member such as a spring **124**, that is configured to rotate the collar **119** when the spring **124** is released. The spring **124** may be a torsion spring. The spring **124** is released when the needle cover **113** reaches a predetermined axial displacement with a release mechanism (not shown). The rotation of the collar **119** causes the plunger **121** to move distally within the syringe **123**, in view of the screw thread **122**, to thereby dispense medicament from the syringe **123** via the needle **117**.

(30) The needle cover **113** is pressed against an injection site, thereby moving the needle cover **113** axially into the body **111** and uncovering the needle **117**. The axial displacement of the needle cover **113** causes the release of the spring **124** which rotates the collar **119**. The rotation of the collar **119** moves the plunger **121** axially within the syringe **123** to dispense the medicament via the needle **117**.

(31) The device **100** is pressed against the injection site **125**, to hold the needle cover **113** at the holding position whilst the medicament is dispensed from the device.

(32) After the medicament has been dispensed, the device **100** is removed from the injection site. The needle cover **113** moves distally under the force of the spring **118** to a locked position. In the locked position, the needle cover **113** covers the distal end **140** of the needle. In the locked position, the needle cover is prevented from moving proximally.

(33) The medicament delivery device **100** is configured to inject greater than 2 ml of medicament and/or the medicament delivery device **100** is configured to inject medicament having a viscosity of greater than 25 cP.
(34) FIG. **3**A shows a simplified view of a medicament delivery device **200**. The features described and/or contemplated in relation to the medicament delivery device **200** may be incorporated in the medicament delivery device **100** as described and/or contemplated above.
(35) The features described and/or contemplated in relation to the medicament delivery device **200** may be incorporated in another medicament delivery device, for example a medicament delivery device having a different mechanism for dispensing medicament to that described in relation to the medicament delivery device **100**, and/or a medicament delivery device which is configured to inject 2 ml or less of medicament and/or a medicament delivery device which is configured to inject medicament having a viscosity of 25 cP or less, and/or a medicament delivery device in which the medicament is contained in a cartridge which is initially separated from the needle when the needle cover is in the initial position.
(36) In FIGS. **3**A to **3**B, **4**A to **4**B, **5**A to **5**B and FIG. **6**, the reference numerals correspond to corresponding features described and/or contemplated above in relation to FIG. **2**.
(37) In FIG. **3**A the medicament delivery device **200** is shown in an initial position. The medicament delivery device **200** has a needle **117** for injecting medicament, a body **111** having a proximal end **130** and a distal end **131**, and a needle cover **113**. The body **111** is configured to be gripped by a user.
(38) The needle cover **113** is proximally movable relative to the body **111** from an initial position to a holding position. In the initial position, as shown for example in FIG. **3**A, the needle cover **113** covers the distal end of the needle. In the holding position, as shown for example in FIG. **4**A, the needle **117** protrudes from the distal end of the needle cover for injecting medicament. When the needle cover moves from the initial position to the holding position it releases a mechanism, such as the mechanism described and/or contemplated in relation to FIG. **2**, which automatically causes medicament to be dispensed from the device **200**. The needle cover **113** is held in the holding position, for example for a predetermined period of time, whilst the medicament is dispensed from the device.
(39) The needle cover **113** is distally movable relative to the body **111** from the holding position to a locked position, as shown for example in FIG. **5**A, in which the needle cover **113** covers the distal end **140** of the needle **117**.
(40) The medicament delivery device **200** has a biasing member such as a spring **118** configured to bias the needle cover **113** distally.
(41) The medicament delivery device **200** has a guide member **141** configured to rotate relative to the needle cover **113**. The guide member **141** has a guide path **154**. The guide path **154** has a first ramp **148**, a second ramp **149** and a locking abutment surface **150**.
(42) The needle cover **113** is rotationally fixed relative to the body **111**. In another embodiment, the needle cover **113** is rotatable relative to the body **111**. The guide member **141** is axially fixed relative to the body **111**. In another embodiment, the guide member **141** is axially movable relative to the body **111**.
(43) The needle cover **113** has a pin **143** configured to engage and move within the guide path **154**. The needle cover has an arm **153**. The arm **153** extends proximally from a sleeve **154** of the needle cover. The pin **143** is provided on the free end of the arm. In another embodiment, the pin **143** is located distally from the free end of the arm **153**.
(44) The medicament delivery device has two arms **153** each of which has a pin **143** located in a separate guide path **154**. In another embodiment just one arm **153** and one guide path **154** may be provided or more than two arms and corresponding guide paths may be provided. The features of just one arm **153** and guide path **154** will be described herein but if other arms and guide paths are present then they may have the same features as described herein in relation to the one arm **153** and

guide path **154**.
(45) In another embodiment, the needle cover **113** has a different construction. For example, the arms may **153** may not be present. The needle cover **113** may be in the form of a circular sleeve. The pin **143** may be provided on an outer surface of the circular sleeve, for example.
(46) The pin **143** is positioned at a first position **145** when the needle cover **113** is in the initial position. The first position **145** is indicated in FIG. **3**B. The first position **145** is in the guide path **154**. The guide path **154** has an axially extending portion **151**. The first position **145** is in the axially extending portion **151**. The axially-extending portion **151** is parallel to the longitudinal axis **144** of the medicament delivery device. In another embodiment, the axially-extending portion **151** is at an acute angle to the longitudinal axis **144** of the medicament delivery device. In another embodiment, the axially-extending portion **151** is not present and the pin **143** engages the first ramp **148** when the needle cover **113** is in the initial position, for example.
(47) In another embodiment, the first position **145** of the pin **143**, when the needle cover is in the initial position, is external to the guide path **154**, for example the pin **143** could drop into guide path **154** after an initial axial movement.
(48) The guide path **154** has a first ramp **148** configured to engage the pin **143** when the needle cover **113** moves axially from the initial position towards the holding position for rotating the guide member **141** relative to the needle cover **113**. The first ramp **148** is located proximally of the first position **145**. The first ramp **148** extends proximally away from the first position **145**. The first ramp **148** rotates the guide member **141** relative to the needle cover **133** when the pin **143** engages and travels along the first ramp **148** and the needle cover **133** moves from the initial position towards the holding position. In another embodiment the first ramp **148** may extend distally from the first position as well as proximally from the first position.
(49) The pin **143** is positioned at a second position **146** in the guide path **154** when the needle cover **133** is in the holding position. The second position is indicated in FIG. **4**B. When the pin **143** is in the second position **146** it engages the first ramp **148**. In another embodiment, the first ramp **148** may connect to an axially-extending portion which is parallel with the longitudinal axis of the device, for example, and the second position **146** may be in the axially-extending portion.
(50) The guide path **150** has a second ramp **149** configured to engage the pin **143** when the needle cover **133** moves axially from the holding position towards the locked position for rotating the guide member **141** relative to the needle cover **113**. The second ramp **149** is located distally of the second position. The second ramp **149** extends distally away from the second position. The second ramp **149** rotates the guide member **141** relative to the needle cover **133** when the pin **143** engages and travels along the second ramp **149** and the needle cover **133** moves from the holding position towards the locked position.
(51) The first ramp **148** is a planar surface which is at an acute angle to the longitudinal axis **144** of the medicament delivery device. The second ramp **149** is a planar surface which is at an acute angle to the longitudinal axis **144** of the medicament delivery device. In another embodiment the first ramp **148** and/or the second ramp **149** comprises a curved surface.
(52) The pin **143** is positioned at a third position **147** in the guide path **154** when the needle cover **113** is in the locked position. The third position **147** is indicated in FIG. **5**B. The guide path **154** has a locking abutment surface **150** located proximally of the third position **147**. The locking abutment surface **150** is configured to engage the pin **143** to prevent the needle cover **113** from moving proximally away from the locked position.
(53) The locking abutment surface **150** extends distally from the third position at an acute angle to the longitudinal axis **144** of the medicament delivery device **200**. In another embodiment, the locking abutment surface **150** may be perpendicular to the longitudinal axis **144** of the medicament delivery device.
(54) If the medicament delivery device **200** did not have a locking functionality for preventing proximal movement of the needle cover **113** after use, then the spring **118** would need to be much

stronger to ensure that the distal end **140** of the needle remains covered once the device has been used. Therefore, having the locking functionality in the device means that the spring **118** can be weaker, which consequently reduces the force that is required to be exerted to hold the device in the holding position.
(55) The guide member **141** has an outer surface **152**. The guide path **154** is recessed within the outer surface **152**. In another embodiment, the guide path **154** may be raised, such as located radially-outwardly, from the outer surface **152**.
(56) The first ramp **148** is configured to rotate the guide member **141** in a first direction when the needle cover **113** moves from the initial position towards the holding position. The second ramp **149** is configured to rotate the guide member **141** in the first direction when the needle cover **113** moves from the holding position towards the locked position. The locking abutment surface **150** is configured to prevent rotation of the guide member **141** in a second direction which is counter to the first direction when the pin **143** is in the third position. In another embodiment, for example if the locking abutment surface **150** is perpendicular to the longitudinal axis **144** then the locking abutment surface does not prevent rotation of the guide member **141** in the second direction.
(57) In another embodiment, the first ramp **148** is configured to rotate the guide member **141** in a first direction when the needle cover **113** moves from the initial position towards the holding position and the second ramp **149** is configured to rotate the guide member in a second direction which is counter to the first direction when the needle cover **113** moves from the holding position towards the locked position.
(58) The guide path **154** provides different paths for the needle cover before and after the medicament has been dispensed from the device.
(59) In the embodiments described and/or contemplated above the needle cover **113** comprises the pin **143** and the guide member **141** comprises the guide path **154**. However, in other embodiments, the guide member **141** comprises the pin **143** and the needle cover **113** comprises the guide path **154**. These other embodiments may have corresponding features to those described and/or contemplated herein in which the needle cover comprises the pin and the guide member comprises the guide path.
(60) The medicament delivery device may have a cap (not shown) which covers the distal end of the needle cover **113** and which must be removed before use.
(61) In use, the needle cover **113** is pressed against an injection site **125**, for example by a user holding the body **111**, to move the needle cover **113** from the initial position to the holding position. The pin **143** is in a first position when the needle cover **113** is in the initial position. When the needle cover moves from the initial position to the holding position then the pin **143** moves in the guide path **154** from the axially-extending portion **151** (if present) to the first ramp **148**. The pin **143** engages the first ramp **148** and rotates the guide member **141** relative to the needle cover **113** and the body **111** when the pin **143** engages the first ramp **148** and travels along the first ramp **148** as the needle cover **113** is moved axially towards the holding position.
(62) When the needle cover **113** is in the holding position then the medicament delivery device **200** is held against the injection site **125** whilst the medicament is dispensed from the device **200**. The pin **143** is in the second position **146** when the needle cover **113** is in the holding position.
(63) After the medicament has been dispensed, the medicament delivery device **200** is removed from the injection site **125**. The needle cover moves distally from the holding position towards the locked position under the force of the biasing member **118**.
(64) When the needle cover moves from the holding position to the locked position then the pin **143** engages the second ramp **149** and rotates the guide member **141** relative to the needle cover **113**.
(65) When the needle cover **113** is in the locked position then the locking abutment surface **150** prevents the needle cover **113** from moving proximally away from the locked position.
(66) FIG. **6** shows a perspective view of a guide path **242** for a medicament delivery device. The

guide path **242** has corresponding features to those described in relation to the guide path **154** but it has a slightly different shape. The reference numerals in FIG. **6** correspond to corresponding features described and/or contemplated above in relation to FIGS. **3**A, **3**B, **4**A, **4**B, **5**A and **5**B.

(67) An example method of using the medicament delivery device **200** will now be described.

(68) The method may include removing the cap (if present) from the medicament delivery device. The method includes pressing the medicament delivery device against the injection site to move the needle cover **113** from an initial position to a holding position for dispensing medicament from the device.

(69) The method then includes holding the medicament delivery device **200** in the holding position whilst medicament is dispensed from the device.

(70) The method then includes removing the medicament delivery device **200** from the injection site. Removing the medicament delivery device from the injection site causes the needle cover **113** of the device to move to a locked position. The needle cover **113** is moved to the locked position under the force of the biasing member **118**. In the locked position proximal movement of the needle cover **113** away from the locked position is blocked by the pin engaging the locking abutment surface **150**.

LIST OF FEATURES

(71) **10**—Device **11**—housing **12**—cap **13**—needle sleeve **17**—needle **20**—distal region **21**—proximal region **22**—button **23**—piston **100**—Device **111**—body **113**—needle cover **117**—needle **118**—spring **119**—collar **120**—distal end of needle cover **121**—plunger **122**—screw thread **123**—syringe **124**—spring **125**—injection site **130**—proximal end of body **131**—distal end of body **140**—distal end of needle **141**—guide member **143**—pin **144**—axis **145**—first position **146**—second position **147**—third position **148**—first ramp **149**—second ramp **150**—locking abutment surface **151**—axially-extending portion **152**—outer surface **153**—arm **154**—guide path **200**—device **242**—guide path

(72) The terms “drug” or “medicament” are used synonymously herein and describe a pharmaceutical formulation containing one or more active pharmaceutical ingredients or pharmaceutically acceptable salts or solvates thereof, and optionally a pharmaceutically acceptable carrier. An active pharmaceutical ingredient (“API”), in the broadest terms, is a chemical structure that has a biological effect on humans or animals. In pharmacology, a drug or medicament is used in the treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being. A drug or medicament may be used for a limited duration, or on a regular basis for chronic disorders.

(73) As described below, a drug or medicament can include at least one API, or combinations thereof, in various types of formulations, for the treatment of one or more diseases. Examples of API may include small molecules having a molecular weight of 500 Da or less; polypeptides, peptides and proteins (e.g., hormones, growth factors, antibodies, antibody fragments, and enzymes); carbohydrates and polysaccharides; and nucleic acids, double or single stranded DNA (including naked and cDNA), RNA, antisense nucleic acids such as antisense DNA and RNA, small interfering RNA (siRNA), ribozymes, genes, and oligonucleotides. Nucleic acids may be incorporated into molecular delivery systems such as vectors, plasmids, or liposomes. Mixtures of one or more drugs are also contemplated.

(74) The drug or medicament may be contained in a primary package or “drug container” adapted for use with a drug delivery device. The drug container may be, e.g., a cartridge, syringe, reservoir, or other solid or flexible vessel configured to provide a suitable chamber for storage (e.g., short- or long-term storage) of one or more drugs. For example, in some instances, the chamber may be designed to store a drug for at least one day (e.g., **1** to at least 30 days). In some instances, the chamber may be designed to store a drug for about 1 month to about 2 years. Storage may occur at room temperature (e.g., about 20° C.), or refrigerated temperatures (e.g., from about −4° C. to about 4° C.). In some instances, the drug container may be or may include a dual-chamber cartridge

configured to store two or more components of the pharmaceutical formulation to-be-administered (e.g., an API and a diluent, or two different drugs) separately, one in each chamber. In such instances, the two chambers of the dual-chamber cartridge may be configured to allow mixing between the two or more components prior to and/or during dispensing into the human or animal body. For example, the two chambers may be configured such that they are in fluid communication with each other (e.g., by way of a conduit between the two chambers) and allow mixing of the two components when desired by a user prior to dispensing. Alternatively or in addition, the two chambers may be configured to allow mixing as the components are being dispensed into the human or animal body.

(75) The drugs or medicaments contained in the drug delivery devices as described herein can be used for the treatment and/or prophylaxis of many different types of medical disorders. Examples of disorders include, e.g., diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy, thromboembolism disorders such as deep vein or pulmonary thromboembolism. Further examples of disorders are acute coronary syndrome (ACS), angina, myocardial infarction, cancer, macular degeneration, inflammation, hay fever, atherosclerosis and/or rheumatoid arthritis. Examples of APIs and drugs are those as described in handbooks such as Rote Liste 2014, for example, without limitation, main groups 12 (anti-diabetic drugs) or 86 (oncology drugs), and Merck Index, 15th edition.

(76) Examples of APIs for the treatment and/or prophylaxis of type 1 or type 2 diabetes mellitus or complications associated with type 1 or type 2 diabetes mellitus include an insulin, e.g., human insulin, or a human insulin analogue or derivative, a glucagon-like peptide (GLP-1), GLP-1 analogues or GLP-1 receptor agonists, or an analogue or derivative thereof, a dipeptidyl peptidase-4 (DPP4) inhibitor, or a pharmaceutically acceptable salt or solvate thereof, or any mixture thereof. As used herein, the terms “analogue” and “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, by deleting and/or exchanging at least one amino acid residue occurring in the naturally occurring peptide and/or by adding at least one amino acid residue. The added and/or exchanged amino acid residue can either be codable amino acid residues or other naturally occurring residues or purely synthetic amino acid residues. Insulin analogues are also referred to as “insulin receptor ligands”. In particular, the term “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, in which one or more organic substituent (e.g., a fatty acid) is bound to one or more of the amino acids. Optionally, one or more amino acids occurring in the naturally occurring peptide may have been deleted and/or replaced by other amino acids, including non-codeable amino acids, or amino acids, including non-codeable, have been added to the naturally occurring peptide. Examples of insulin analogues are Gly (A21), Arg (B31), Arg (B32) human insulin (insulin glargine); Lys (B3), Glu (B29) human insulin (insulin glulisine); Lys (B28), Pro (B29) human insulin (insulin lispro); Asp (B28) human insulin (insulin aspart); human insulin, wherein proline in position B28 is replaced by Asp, Lys, Leu, Val or Ala and wherein in position B29 Lys may be replaced by Pro; Ala (B26) human insulin; Des (B28-B30) human insulin; Des (B27) human insulin and Des (B30) human insulin.

(77) Examples of insulin derivatives are, for example, B29-N-myristoyl-des (B30) human insulin, Lys (B29) (N-tetradecanoyl)-des (B30) human insulin (insulin detemir, Levemir®); B29-N-palmitoyl-des (B30) human insulin; B29-N-myristoyl human insulin; B29-N-palmitoyl human insulin; B28-N-myristoyl LysB28ProB29 human insulin; B28-N-palmitoyl-LysB28ProB29 human insulin; B30-N-myristoyl-ThrB29LysB30 human insulin; B30-N-palmitoyl-ThrB29LysB30 human insulin; B29-N-(N-palmitoyl-gamma-glutamyl)-des (B30) human insulin, B29-N-omega-carboxypentadecanoyl-gamma-L-glutamyl-des (B30) human insulin (insulin degludec, Tresiba®); B29-N-(N-lithocholyl-gamma-glutamyl)-des (B30) human insulin; B29-N-(ω-carboxyheptadecanoyl)-des (B30) human insulin and B29-N-(ω-carboxyheptadecanoyl) human

insulin.
(78) Examples of GLP-1, GLP-1 analogues and GLP-1 receptor agonists are, for example, Lixisenatide (Lyxumia®), Exenatide (Exendin-4, Byetta®, Bydureon®, a 39 amino acid peptide which is produced by the salivary glands of the Gila monster), Liraglutide (Victoza®), Semaglutide, Taspoglutide, Albiglutide (Syncria®), Dulaglutide (Trulicity®), rExendin-4, CJC-1134-PC, PB-1023, TTP-054, Langlenatide/HM-11260C (Efpeglenatide), HM-15211, CM-3, GLP-1 Eligen, ORMD-0901, NN-9423, NN-9709, NN-9924, NN-9926, NN-9927, Nodexen, Viador-GLP-1, CVX-096, ZYOG-1, ZYD-1, GSK-2374697, DA-3091 March-701, MAR709, ZP-2929, ZP-3022, ZP-DI-70, TT-401 (Pegapamodtide), BHM-034. MOD-6030, CAM-2036, DA-15864, ARI-2651, ARI-2255, Tirzepatide (LY3298176), Bamadutide (SAR425899), Exenatide-XTEN and Glucagon-Xten. An example of an oligonucleotide is, for example: mipomersen sodium (Kynamro®), a caperturesterol-reducing antisense therapeutic for the treatment of familial hypercaperturesterolemia or RG012 for the treatment of Alport syndrom. Examples of DPP4 inhibitors are Linagliptin, Vildagliptin, Sitagliptin, Denagliptin, Saxagliptin, Berberine.
(79) Examples of hormones include hypophysis hormones or hypothalamus hormones or regulatory active peptides and their antagonists, such as Gonadotropine (Follitropin, Lutropin, Choriongonadotropin, Menotropin), Somatropine (Somatropin), Desmopressin, Terlipressin, Gonadorelin, Triptorelin, Leuprorelin, Buserelin, Nafarelin, and Goserelin.
(80) Examples of polysaccharides include a glucosaminoglycane, a hyaluronic acid, a heparin, a low molecular weight heparin or an ultra-low molecular weight heparin or a derivative thereof, or a sulphated polysaccharide, e.g. a poly-sulphated form of the above-mentioned polysaccharides, and/or a pharmaceutically acceptable salt thereof. An example of a pharmaceutically acceptable salt of a poly-sulphated low molecular weight heparin is enoxaparin sodium. An example of a hyaluronic acid derivative is Hylan G-F 20 (Synvisc®), a sodium hyaluronate.
(81) The term “antibody”, as used herein, refers to an immunoglobulin molecule or an antigen-binding portion thereof. Examples of antigen-binding portions of immunoglobulin molecules include F (ab) and F(ab′)2 fragments, which retain the ability to bind antigen. The antibody can be polyclonal, monoclonal, recombinant, chimeric, de-immunized or humanized, fully human, non-human, (e.g., murine), or single chain antibody. In some embodiments, the antibody has effector function and can fix complement. In some embodiments, the antibody has reduced or no ability to bind an Fc receptor. For example, the antibody can be an isotype or subtype, an antibody fragment or mutant, which does not support binding to an Fc receptor, e.g., it has a mutagenized or deleted Fc receptor binding region. The term antibody also includes an antigen-binding molecule based on tetravalent bispecific tandem immunoglobulins (TBTI) and/or a dual variable region antibody-like binding protein having cross-over binding region orientation (CODV).
(82) The terms “fragment” or “antibody fragment” refer to a polypeptide derived from an antibody polypeptide molecule (e.g., an antibody heavy and/or light chain polypeptide) that does not comprise a full-length antibody polypeptide, but that still comprises at least a portion of a full-length antibody polypeptide that is capable of binding to an antigen. Antibody fragments can comprise a cleaved portion of a full length antibody polypeptide, although the term is not limited to such cleaved fragments. Antibody fragments that are useful in the present invention include, for example, Fab fragments, F(ab′)2 fragments, scFv (single-chain Fv) fragments, linear antibodies, monospecific or multispecific antibody fragments such as bispecific, trispecific, tetraspecific and multispecific antibodies (e.g., diabodies, triabodies, tetrabodies), monovalent or multivalent antibody fragments such as bivalent, trivalent, tetravalent and multivalent antibodies, minibodies, chelating recombinant antibodies, tribodies or bibodies, intrabodies, nanobodies, small modular immunopharmaceuticals (SMIP), binding-domain immunoglobulin fusion proteins, camelized antibodies, and VHH containing antibodies. Additional examples of antigen-binding antibody fragments are known in the art. The terms “Complementarity-determining region” or “CDR” refer to short polypeptide sequences within the variable region of both heavy and light chain

polypeptides that are primarily responsible for mediating specific antigen recognition. The term "framework region" refers to amino acid sequences within the variable region of both heavy and light chain polypeptides that are not CDR sequences, and are primarily responsible for maintaining correct positioning of the CDR sequences to permit antigen binding. Although the framework regions themselves typically do not directly participate in antigen binding, as is known in the art, certain residues within the framework regions of certain antibodies can directly participate in antigen binding or can affect the ability of one or more amino acids in CDRs to interact with antigen.

(83) Examples of antibodies are anti PCSK-9 mAb (e.g., Alirocumab), anti IL-6 mAb (e.g., Sarilumab), and anti IL-4 mAb (e.g., Dupilumab).

(84) Pharmaceutically acceptable salts of any API described herein are also contemplated for use in a drug or medicament in a drug delivery device. Pharmaceutically acceptable salts are for example acid addition salts and basic salts.

(85) Those of skill in the art will understand that modifications (additions and/or removals) of various components of the APIs, formulations, apparatuses, methods, systems and embodiments described herein may be made without departing from the full scope and spirit of the present invention, which encompass such modifications and any and all equivalents thereof.

(86) An example drug delivery device may involve a needle-based injection system as described in Table 1 of section 5.2 of ISO 11608-1: 2014 (E). As described in ISO 11608-1: 2014 (E), needle-based injection systems may be broadly distinguished into multi-dose container systems and single-dose (with partial or full evacuation) container systems. The container may be a replaceable container or an integrated non-replaceable container.

(87) As further described in ISO 11608-1: 2014 (E), a multi-dose container system may involve a needle-based injection device with a replaceable container. In such a system, each container holds multiple doses, the size of which may be fixed or variable (pre-set by the user). Another multi-dose container system may involve a needle-based injection device with an integrated non-replaceable container. In such a system, each container holds multiple doses, the size of which may be fixed or variable (pre-set by the user).

(88) As further described in ISO 11608-1: 2014 (E), a single-dose container system may involve a needle-based injection device with a replaceable container. In one example for such a system, each container holds a single dose, whereby the entire deliverable volume is expelled (full evacuation). In a further example, each container holds a single dose, whereby a portion of the deliverable volume is expelled (partial evacuation). As also described in ISO 11608-1: 2014 (E), a single-dose container system may involve a needle-based injection device with an integrated non-replaceable container. In one example for such a system, each container holds a single dose, whereby the entire deliverable volume is expelled (full evacuation). In a further example, each container holds a single dose, whereby a portion of the deliverable volume is expelled (partial evacuation).

Claims

1. A medicament delivery device comprising: a needle for injecting medicament; a body having a proximal end and a distal end; a needle cover, wherein the needle cover is proximally movable relative to the body from an initial position to a holding position in which the needle protrudes from the distal end of the needle cover for injecting medicament, and wherein the needle cover is distally movable relative to the body from the holding position to a locked position in which the needle cover covers the distal end of the needle; a biasing member configured to bias the needle cover distally; and a guide member configured to rotate relative to the needle cover, wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path for guiding the pin, wherein the guide path comprises a first ramp, a second ramp and a locking abutment surface, wherein the pin is positioned at a first

position when the needle cover is in the initial position, wherein the first position is external to the guide path, and wherein the first ramp is configured to engage the pin when the needle cover moves axially from the initial position towards the holding position for rotating the guide member relative to the needle cover, wherein the pin is positioned at a second position when the needle cover is in the holding position, wherein the second ramp is configured to engage the pin when the needle cover moves axially from the holding position towards the locked position for rotating the guide member relative to the needle cover, and wherein the pin is positioned at a third position when the needle cover is in the locked position, wherein the locking abutment surface is radially fixed relative to the guide member and configured to engage the pin to prevent the needle cover from moving proximally away from the locked position.

2. The medicament delivery device according to claim 1, wherein the locking abutment surface comprises a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device.

3. The medicament delivery device according to claim 1, wherein when the pin is in the second position it engages the first ramp.

4. The medicament delivery device according to claim 1, wherein the first position is in the guide path, and optionally wherein the guide path comprises an axially extending portion, wherein the first position is in the axially extending portion, and optionally wherein the axially extending portion is parallel with the longitudinal axis of the medicament delivery device.

5. The medicament delivery device according to claim 1, wherein the needle cover is rotationally fixed relative to the body and/or wherein the guide member is axially fixed relative to the body.

6. The medicament delivery device according to claim 1, wherein the first ramp comprises a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device and/or wherein the second ramp comprises a planar surface which is at an acute angle to the longitudinal axis of the medicament delivery device.

7. The medicament delivery device according to claim 1, wherein the body is configured to be gripped by a user.

8. The medicament delivery device according to claim 1, wherein the needle cover comprises the pin and the guide member comprises the guide path and/or wherein in the initial position the needle cover covers the distal end of the needle.

9. The medicament delivery device according to claim 8, wherein the guide member comprises an outer surface, and wherein the guide path is recessed within the outer surface.

10. The medicament delivery device according to claim 8, wherein the needle cover comprises an arm, wherein the pin is provided on the arm, and optionally wherein the pin is provided on the free end of the arm.

11. The medicament delivery device according to claim 1, wherein the first ramp is configured to rotate the guide member in a first direction when the needle cover moves axially away from the initial position towards the holding position, and wherein the second ramp is configured to rotate the guide member in the first direction when the needle cover moves axially from the holding position towards the locked position, and optionally wherein the locking abutment surface is configured to prevent rotation of the guide member in a second direction which is counter to the first direction when the pin is in the third position.

12. The medicament delivery device according to claim 1, wherein the medicament delivery device is configured to inject greater than 2 ml of medicament and/or wherein the medicament delivery device is configured to inject medicament having a viscosity of greater than 25 cP.

13. The medicament delivery device according to claim 1, wherein the device comprises a mechanism which automatically causes medicament to be dispensed from the device, wherein when the needle cover moves from the initial position towards the holding position it releases the mechanism and medicament is automatically dispensed from the device.

14. The medicament delivery device according to claim 1, wherein the device comprises the

medicament.

15. A method of locking a medicament delivery device after medicament has been dispensed from the medicament delivery device, the method comprising moving a needle cover of the device to a locked position in which the needle cover covers the distal end of a needle of the device, wherein the device comprises a guide member configured to rotate relative to the needle cover, and wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path for guiding the pin, wherein the pin is positioned at a first position when the needle cover is in an initial position, wherein the first position is external to the guide path, wherein the guide path comprises a locking abutment surface that is radially fixed relative to the guide member, and wherein when the needle cover is in the locked position then proximal movement of the needle cover away from the locked position is blocked by the pin engaging the locking abutment surface.

16. A method of using a medicament delivery device, the method comprising removing the medicament delivery device from an injection site, wherein removing the medicament delivery device from the injection site causes a needle cover of the device to move to a locked position in which the needle cover covers the distal end of a needle of the device, wherein the device comprises a guide member configured to rotate relative to the needle cover, and wherein one of the guide member and the needle cover comprises a pin and the other of the guide member and the needle cover comprises a guide path comprising a locking abutment surface that is radially fixed relative to the guide member, wherein the pin is positioned at a first position when the needle cover is in an initial position, wherein the first position is external to the guide path, and wherein when the needle cover is in the locked position then proximal movement of the needle cover away from the locked position is blocked by the pin engaging the locking abutment surface.

17. The method according to claim 16, further comprising the preceding step of pressing the medicament delivery device against the injection site to move the needle cover from the initial position to a holding position for dispensing medicament from the device.

18. The method according to claim 17, further comprising holding the medicament delivery device in the holding position whilst medicament is dispensed from the device.

19. The method according to claim 17, wherein the guide path further comprises a first ramp which engages the pin when the needle cover moves axially from the initial position towards the holding position for rotating the guide member relative to the needle cover.

20. The method according to claim 16, wherein the guide path further comprises a second ramp which engages the pin when the needle cover moves from the holding position towards the locked position for rotating the guide member relative to the needle cover.
